

A novel approach to determine the time step in hydrodynamic simulations based on deep learning

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Abstract

This paper puts forward a novel approach for determining the time step by leveraging deep learning methods within hydrodynamic simulations. In traditional hydraulic simulations, the size of the time step was determined by the Courant-Friedrich-Lewy (CFL) condition, which represents the minimum across the entire domain and entails a certain degree of "redundancy". In this study, a deep learning model named SwinT, which is based on the Swin Transformer [1] is employed to undertake the time step determination task. The training data for SwinT is sourced from TRITON [2]. Eventually, the model is coupled with TRITON via ONNX to jointly perform the time step prediction task. The experimental results demonstrate that by adjusting the hyperparameters of SwinT, the coupled model can handle diverse situations. High accuracy can be achieved by compromising a degree of speed, although it will remain faster than the original model. For instance, in Test 1, the error of the coupled model with respect to the theoretical value is significantly smaller than that of the original model. Conversely, prioritizing speed necessitates a compromise in accuracy, resulting in a final error within the millimeter range. For example, in Test 2, when the value of q is set to 0.85, the simulation time is reduced by 22.29%, while the error in water depth is only within the millimeter range.

CCS Concepts

• **Computing methodologies** → Modeling and simulation; Model development and analysis.

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CAICE 2025, Heifei, China
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ACM ISBN 979-8-4007-1264-7/2025/01
<https://doi.org/10.1145/3727648.3727691>

Keywords

hydrodynamic simulation, CFL condition, coupled model, deep learning, TRITON

ACM Reference Format:

Wensheng Tang, Jieru Wei, Jiandong Shang, Jichang Yang, and Junjie Chen. 2025. A novel approach to determine the time step in hydrodynamic simulations based on deep learning. In *The 4th International Conference on Computer, Artificial Intelligence and Control Engineering (CAICE 2025)*, January 10–12, 2025, Heifei, China. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3727648.3727691>

1 INTRODUCTION

Floods are likely to be among the most severe and recurrent natural disasters that human societies face, as they typically arise swiftly and ominously. Throughout human history, numerous efforts have been made to comprehend, assess, and predict flood events and their consequences. To this end, the flood model has been developed.

Flood models can be categorized into hydrological models and hydrodynamic models [3]. Hydrological models employ a simplified version of the mass or momentum equation to circumvent the direct solution of the nonlinear hydrodynamic processes [4]. Conversely, hydrodynamic models are formulated to address fully nonlinear systems of shallow water equations [5]. This enables hydrodynamic models to capture detailed hydrodynamic properties, such as flood arrival time, inundation area, water depth, and flow velocity [6]. Using rainfall as the input, Liang [7] predicts the entire process of fluvial flooding, ranging from source (rainfall) to impact (inundation), over a large catchment by utilizing a single high-performance hydrodynamic model. The PRIMO [8] (Parallel Raster Inundation Model) tackles the shallow water equations on an upscaled grid which is significantly coarser than the underlying Digital Topographic Model (DTM) input. Moreover, it adopts a sub-grid modelling approach so that the solution can take advantage of DTM-scale topographic data. With the application of MPI (Message Passing Interface) and CUDA (Compute Unified Device Architecture), four different test cases were simulated on the Summit supercomputer at Oak Ridge National Laboratory. The

simulation was scaled up to 128 nodes (768 GPUs), achieving a maximum speedup of 98.2x over 1 GPU, as reported in [9].

The 2D flood models are capable of accurately simulating the dynamics of overland flows. They are significantly less dependent on parameterization and can reliably predict rainfall-runoff processes and subsequent flood propagation [10]. Regrettably, when these models are employed to simulate large-scale, long-term events with high-resolution computational meshes, they prove to be computationally expensive [11]. One effective approach is adaptive mesh refinement (AMR), which recursively refines the mesh within specific parts of the domain and adaptively updates the refinement as the simulation proceeds [12]. Another factor contributing to the cost lies in the Courant-Friedrichs-Lewy (CFL) constraint, and (in most cases) the adopted global minimum time step for variable updating [13]. In (Peng Hu) [14], a hybrid local time step/global maximum time step (LTS/GMaTS) approach is put forward to enable computationally efficient modeling of hydro-sediment-morphodynamic processes. A GPU-accelerated finite volume shallow water model, based on the Local Time Step (LTS) method, has been extended for rainfall-runoff simulations through coupling with the Green-Ampt model, similar to the approach described in Wu [15]. All of these endeavors have led to significant acceleration ratios.

Without advancements in hardware, conventional hydraulic simulations struggle to make significant improvements in speed and accuracy. In recent years, deep learning has experienced rapid advancement and achieved significant outcomes in hydraulic simulation. Utilizing convolutional neural networks and fully connected neural networks, with the two-dimensional shallow water equations and initial and boundary conditions functioning as loss functions during training, the authors attained superior results relative to conventional finite volume-based models, as demonstrated in [16]. Nevertheless, the question of how A can be more effectively coupled with B remains a matter worthy of in-depth consideration.

In this article, a novel deep learning method for determining the time step in hydraulic simulations is put forward. Specifically, based on the Swin Transformer, the SwinT is trained using the data generated by TRITON. Moreover, the hyperparameter of the percentage loss function is utilized to govern the trade-off between accuracy and time. As a result, the final coupled model can attain either elevated simulation speed or enhanced simulation accuracy.

The remainder of this paper is structured as follows. Firstly, TRITON, SwinT, and the coupling model are elaborated on in Section 2. Subsequently, in Section 3, the accuracy and computational performance of the proposed coupling model are examined with respect to the test case. Finally, the main outcomes of the study are discussed and summarized in Section 4.

2 METHODS

2.1 TRITON

The two-dimensional shallow water equations describe the depth-averaged conservation of mass and momentum in the x and y directions within the space. Their conservative form is presented in Eqs (1-3) as follows:

$$\frac{\partial U}{\partial t} + \frac{\partial F}{\partial x} + \frac{\partial G}{\partial y} = S_r + S_b + S_f \quad (1)$$

$$U = \begin{pmatrix} h \\ q_x \\ q_y \end{pmatrix} \quad F = \begin{pmatrix} q_x \\ \frac{q_x^2}{h} + \frac{gh^2}{2} \\ \frac{q_x q_y}{h} \end{pmatrix} \quad G = \begin{pmatrix} q_y \\ \frac{q_x q_y}{h} \\ \frac{q_y^2}{h} + \frac{gh^2}{2} \end{pmatrix} \quad (2)$$

$$S_r = \begin{pmatrix} r \\ 0 \\ 0 \end{pmatrix} \quad S_b = \begin{pmatrix} 0 \\ -gh \frac{\partial Z}{\partial x} \\ -gh \frac{\partial Z}{\partial y} \end{pmatrix} \quad S_f = \begin{pmatrix} 0 \\ -\frac{gn^2}{h^{1/3}} q_x \sqrt{q_x^2 + q_y^2} \\ -\frac{gn^2}{h^{1/3}} q_y \sqrt{q_x^2 + q_y^2} \end{pmatrix} \quad (3)$$

here, vector U stands for the unknown variables, including water depth h , q_x and q_y , g is the gravity acceleration and the source terms, and F and G denote the fluxes of these conserved variables. Eqs. (1-3) also encompasses other source terms including runoff S_r based on the runoff rate r , bed slope S_b based on the slope gradient Z , and friction S_f modeled by the Gauckler-Manning's law with regard to the Manning's roughness coefficient n . Here, infiltration, abstraction, and evapotranspiration are all factors that contribute to the runoff.

To solve Eq.(1), a finite volume upwind explicit scheme is employed within a Cartesian grid. The global minimum time step, which is determined based on the CFL condition, is used to update the variables like water depth and discharges similar to Eq.(4). The CFL condition defines the relationship between the time step (Δt) and the space step (Δx). Specifically, it stipulates that information travels no more than the length of a grid cell within a time step.

$$\Delta t_{min} = C_r \min_i \left\{ \frac{\Delta x}{|u_i| + \sqrt{gh_i}, |v_i| + \sqrt{gh_i}} \right\} \quad (4)$$

where index i loops through all the grid cells, C_r is the Courant number, h_i , u_i , v_i denote the water depth, the flow velocity in the x -direction and the flow velocity in the y -direction of the cell i respectively, Δx stands for the cell size, and h_{thr} is the water depth threshold, which is set to $10e-5$ m. Subsequently, the global time step is calculated as the minimum value among all Δt_i . For more information, please refer to Morales-Hernández [2].

2.2 Coupling model

The coupling model comprises SwinT and TRITON. Upon completion of training, SwinT is exported in the ONNX format. Thereafter, it is integrated into TRITON through ONNX Runtime to facilitate collaborative task completion.

SwinT is founded on Swin_T, which is part of the Swin Transformer, along with the MLP layer. The Swin_T output and several other indicators, like rainfall coefficient and river runoff coefficient, are concatenated via the MLP layer, to obtain the coupling model's prediction results. The architecture of SwinT is illustrated in Figure 1. Swin_T was selected as the backbone network for SwinT given its robust capability in handling matrix data which makes it highly suitable for processing DEM, water depth, and flow rate.

Swin_T comprises four stages, with the respective numbers of Swin Transformer blocks in each stage being (2, 2, 6, 2). Its input comprises matrix data representing the simulation outcomes from the final time step of TRITON. Subsequently, the Swin_T output along with some other indicators are fed into the MLP layer to obtain the final result.

The data within the dataset utilized for training SwinT is generated by TRITON. This dataset is created by preprocessing a total of

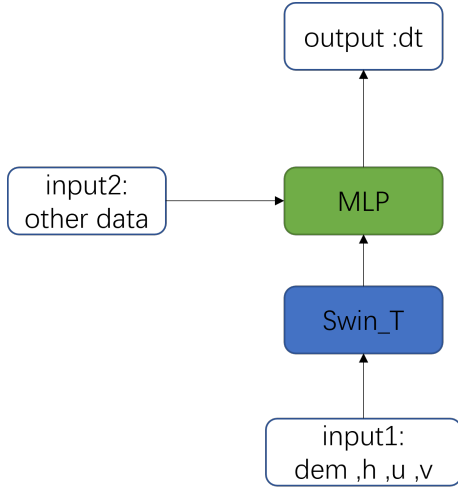


Figure 1: SwinT architecture.

8000 data points. The data allocation for training, validation, and testing is 8:1:1.

The hyperparameters employed in the model are as follows: window_size = 7, embed_dim = 96, depths = (2, 2, 6, 2), num_heads = (3, 6, 12, 24). The parameters used in the training process included: Epochs = 10, batch_size = 8, Optimizer = Adamw/Adam, and Loss Function = quantile loss. It should be noted that the Loss Function parameters can significantly influence the performance of both SwinT and the coupling model. Additional information is presented in the subsequent section.

3 TEST CASE AND RESULTS

3.1 Test 1 paraboloid bed topography without friction

The GPU employed in this experiment is the GeForce RTX 3080. The cuda version is 12.3, while the cudnn version is 9.1.1.17. Additionally, we used the ONNX Runtime 1.18.1, and the Ubuntu 20.04.6 LTS operating system.

The loss function adopted for the training stage of SwinT is the fractional quantile loss function. Its implementation is demonstrated by Eq. 5, where q is set as 0.5.

$$L_q = \max(q(y_n - y_e), (q - 1)(y_n - y_e)) \quad (5)$$

here, y_n and y_e represent the simulated solution and the exact solution, respectively. In this experiment, a frictionless paraboloid is utilized as a digital elevation model within the range of $[0, L] \times [0, L]$. The relationship between the elevation z , x , and y is presented in Eq. 6). The initial condition is defined as a plane with an initial velocity in the y direction. Given that the plane is frictionless, the solution is periodic. The exact solution is provided by Eq. (7-9).

$$z(x, y) = -h_0 \left(1 - \frac{\left(x - \frac{L}{2}\right)^2 + \left(y - \frac{L}{2}\right)^2}{a^2} \right) \quad (6)$$

$$h(x, y, t) = \frac{\eta h_0}{a^2} \left(2 \left(x - \frac{L}{2} \right) \cos(\omega t) + 2 \left(y - \frac{L}{2} \right) \sin(\omega t) - \eta \right) - z(x, y) \quad (7)$$

$$u(x, y, t) = -\eta \omega \sin(\omega t) \quad (8)$$

$$v(x, y, t) = \eta \omega \cos(\omega t) \quad (9)$$

For this experiment, we set $a = 1\text{m}$, $h_0 = 0.1\text{m}$, $\eta = 0.5$, $L = 4\text{m}$, $T = \frac{6\pi}{\omega}$.

Four sets of experiments are conducted herein, with resolutions of 0.04m, 0.02m, 0.01m, and 0.005m, respectively. The aim is to investigate the difference between the coupling model solution and the exact solution at times T , $2T$ and $3T$. Here, we utilize L_1 , L_2 , L_∞ to represent the error, and the specific calculation formulae are presented in Eq. (10-12).

$$L_1(y_{\Delta x}) = \frac{1}{N} \sum_{i=1}^N |(y_n - y_e)_i| \quad (10)$$

$$L_2(y_{\Delta x}) = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_n - y_e)_i^2} \quad (11)$$

$$L_\infty(y_{\Delta x}) = \max_i |(y_n - y_e)_i| \quad (12)$$

where N represents the number of grids, y_n denotes the simulated solution and y_e represents the exact solution.

The coupling model errors are presented in Table 1, while the TRITON errors are presented in Table 2. The water depth conditions at the cross section $x = L/2$ at T , $2T$ and $3T$ along with the details of the left and right dry/wet boundaries at $3T$ are illustrated in Figure 2.

The values of L_1 , L_2 , L_∞ for both the simulation results and the theoretical results gradually decline as the grid resolution increases. Moreover, the smallest L-paradigm numbers are all observed for a $\Delta x = 0.005\text{m}$. The L -paradigm numbers from both the simulation and theoretical outcomes of the coupled model are inferior to those from the simulation and theoretical outcomes of the TRITON model. This demonstrates that the method proposed in this paper is superior to the time-step determination method in TRITON in terms of accuracy when $q = 0.5$. As illustrated in Figure 2, the errors progressively accumulate over time, as an unavoidable occurrence.

3.2 Test 2 a real flood event

The hardware and software utilized in this experiment are identical to those used in test1. This test aimed to simulate a flood event by TRITON. The test site is located upstream of Allatoona Reservoir in the northwest of Georgia, US.

The loss function adopted for training the network is the fractional loss function. Using the fractional loss function may not be the best option, based solely on SwinT's performance. However, after coupling the TRITON-trained network, the resulting coupling model demonstrated a good performance. Furthermore, its parameters can be changed to allow the model to operate more quickly or accurately. This is the exact reason we chose the fractional loss function.

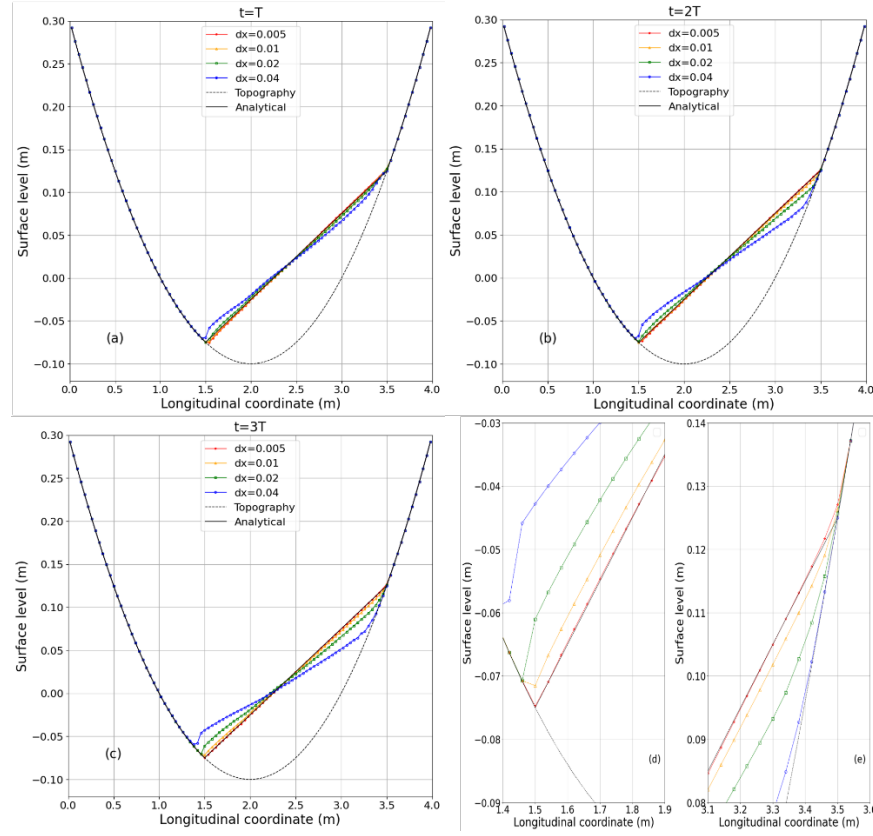
We compared the mean absolute error (MAE), mean square error (MSE), and root mean square error (RMSE) of the simulation results (water depth, flow velocity) in comparison to the TRITON simulation results. This comparison was made when coupling was performed with models trained with varying scores.

Table 1: Coupling model errors.

Δx	L_1	L_2	L_∞
0.04	3.90160e-03	9.51772e-03	4.5371e-02
0.02	2.04638e-03	5.10967e-03	2.3354e-02
0.01	9.61793e-04	2.43575e-03	1.0730e-02
0.005	4.42095e-04	1.11695e-03	5.2437e-03

Table 2: TRITON errors.

Δx	L_1	L_2	L_∞
0.04	3.943e-03	9.612e-03	4.630e-02
0.02	2.065e-03	5.134e-03	2.405e-02
0.01	9.724e-04	2.457e-03	1.090e-02
0.005	4.433e-04	1.117e-03	5.311e-03

**Figure 2: the exact and numerical solutions for different grid resolutions at (a) $t = 1T$, (b) $t = 2T$ and (c) $t = 3T$. Panels (d) and (e) illustrate a magnified view of the left and right wet/dry interfaces, respectively, at $t = 3T$.**

When employing the Adam optimizer and training the model with fractional loss functions at varying values of q , the final simulation durations of the coupled models differ, as illustrated in Table 3. Starting from 0.5, the time consumed by the simulation diminishes as q increases. TRITON took 638.8s to simulate this 5-day flood

event. Using the Adam optimized, the coupling model could reduce the simulation time by up to 22.29% at $q = 0.85$. Using the Adamw optimizer, the coupling model took merely 481.8s to carry out the simulation, which is 24.58% less than that of the TRITON model.

Table 3: Total simulation duration with the Adam optimizer and the percentage decrease in time relative to the original model.

q	0.5	0.6	0.7	0.75	0.8	0.85
time/s	626.9	608	584.8	548	519.6	496.4
reduction/%	1.86	4.82	8.45	14.21	18.66	22.29

Table 4: Simulation accuracy when $q = 0.5$.

	MAE	MSE	RMSE
H	5.44683e-06	1.17663e-07	3.4302e-04
QX	1.80232e-06	2.10843e-09	4.59177e-05
QY	1.7884e-06	2.5013e-09	5.0013e-05

Table 5: Simulation accuracy when $q = 0.6$.

	MAE	MSE	RMSE
H	6.45597e-06	6.15022e-08	2.47996e-04
QX	2.28632e-06	4.8386e-09	6.95601e-05
QY	2.1577e-06	3.17299e-09	5.63293e-05

Table 6: Simulation accuracy when $q = 0.7$.

	MAE	MSE	RMSE
H	8.1926e-06	8.43279e-09	9.18302e-05
QX	2.55549e-06	3.83432e-09	6.19219e-05
QY	2.55122e-06	6.31899e-09	7.94921e-05

The accuracies of different fractional simulations utilizing the Adam optimizer are displayed in Table 4, Table 5, Table 6, Table 7, Table 8 and Table 9. Here, H (m), QX (mm/s), and QY (mm/s) represent the product of the final water depth, velocity and water depth in the X and Y directions, respectively. Meanwhile, MAE, MSE, and RMSE denote the mean absolute error, mean square error, and root-mean-square error of the coupling model and TRITON, respectively.

The simulation accuracies experience a sudden alteration at $q = 0.75$. Using MAE as an example, when $q < 0.75$, the H error, QX, and QY are all of the order of 10^{-6} . However, when $q > 0.75$, the H error is of the order of 10^{-3} , the QX error is of the order of 10^{-4} , and the QY error is of the order of 10^{-4} .

4 CONCLUSIONS AND DISCUSSION

In this research, we have put forward a time-step determination method based on deep learning. The backbone network of SwinT is the Swin_T. The output of Swin_T, along with several other matrices, is fed into the subsequent MLP layer to obtain the final result. The training data for SwinT is obtained from TRITON. Following post-processing, the raw data is converted into a dataset comprising 8,000 data points for training, validation, and testing. Finally, the coupling is achieved by importing SwinT into TRITON via ONNX. The experimental results indicate that the method presented in this

paper effectively determines the time step, substituting the CFL condition, with high accuracy and increased speed.

Despite the coupling model's considerable advantage, its accuracy remains inadequate. This is due to a number of different situations. Firstly, hydrodynamic simulations encompass numerous intricate physical processes, such as turbulent flow, boundary layer effects, and the interactions between different media. These physical phenomena are intertwined, forming a highly nonlinear and complex system. During the training process of the SwinT model, it is extremely challenging to accurately capture the characteristics of these physical processes. With limited data, the SwinT model struggles to learn the complete and accurate physical laws. In the prediction phase, this insufficient understanding of the physical processes causes the time step determined by SwinT to deviate from the optimal one required by the actual physical process. As the simulation progresses, errors accumulate gradually, ultimately leading to significant errors in the results. Data uncertainty is another crucial factor affecting the accuracy of the coupled model. In hydrodynamic simulations, the data sources for determining the time step are extensive and complex, and issues like data missing and insufficient representativeness are prevalent. The data with errors entering the training stage of SwinT will disrupt the model's learning of the true physical relationships, thus having a substantial impact on the simulation results. If the data of certain key influencing factors are missing, SwinT cannot obtain complete information

Table 7: Simulation accuracy when $q = 0.75$.

	MAE	MSE	RMSE
H	1.20092e-05	9.46687e-09	9.72978e-05
QX	3.03817e-06	5.65297e-09	7.51862e-05
QY	3.01968e-06	8.39499e-09	9.16242e-05

Table 8: Simulation accuracy when $q = 0.8$.

	MAE	MSE	RMSE
H	1.06403e-03	9.28432e-04	3.04702e-02
QX	3.78366e-04	1.70432e-03	4.12834e-02
QY	3.81888e-04	1.90526e-03	4.36429e-02

Table 9: Simulation accuracy when $q = 0.85$.

	MAE	MSE	RMSE
H	2.99713e-03	6.79825e-03	8.24515e-02
QX	6.14515e-04	3.00131e-03	5.47842e-02
QY	6.29713e-04	3.13452e-03	5.59868e-02

during training and will construct an incomplete feature representation. During prediction, when faced with these insufficiently learned data situations, the accuracy of SwinT will inevitably be affected. Moreover, if the training data fails to comprehensively represent various complex hydrodynamic scenarios, when SwinT encounters new situations that are not fully reflected in the training data, it will be difficult for the model to make accurate predictions, ultimately affecting the accuracy of the coupled model.

Therefore, for the future enhancement of this method, it is essential to select more appropriate parameters and a wider variety of datasets to train the model. For example, commonly utilized loss functions in regression models could include Mean Squared Error (MSE), Huber loss, and Log-Cosh. When constructing the dataset, data from multiple flood events should be selected to increase both the diversity of the dataset and the number of samples.

The method put forward in this paper only accomplishes one task within the entire process of hydraulics simulation based on deep learning, namely, the time step determination task. In the future, we plan to utilize the deep learning method to complete the entire simulation task. Our objective is to enable the model to directly produce flood data, including water depth, after a specified duration. This will require not only the encoder structure of Swin_T but also the subsequent decoder structure. Evidently, it is going to be a highly challenging task.

In the realm of hydrodynamic simulations, deep learning exhibits significant potential benefits. It is adept at handling complex non-linear relationships, precisely discerning features of highly non-linear physical processes such as turbulent flow and boundary-layer effects, thus effectively overcoming the limitations of traditional approaches in describing these phenomena. Deep learning's successful applications in other fields offer valuable references for hydrodynamic simulation research. For example, convolutional

neural networks, by efficiently processing extensive meteorological data, have achieved high-precision early predictions of the El Niño phenomenon. In power equipment condition monitoring, deep-learning-based switchgear partial-discharge detection and fault-identification technologies have remarkably improved detection accuracy and intelligence. These cases validate the feasibility and broad prospects of incorporating deep learning into hydrodynamic simulations. This is likely to drive innovative breakthroughs, opening up new avenues for related research and applications.

Acknowledgments

I am profoundly grateful to my supervisor, Professor Shang Jiandong, and Dr. Wei Jieru, for their invaluable guidance and unwavering support throughout this research endeavor. Their profound knowledge and perceptive suggestions have played a pivotal role in molding this work and assisting me in surmounting numerous hurdles. I would like to convey my heartfelt gratitude to my fellow classmates, Yang Jichang, Chen Junjie, and others. They offered invigorating discussions and shared their ideas and experiences, which expanded my research horizons and made the research process all the more enjoyable. My sincere appreciation also extends to the participants of the study, who selflessly dedicated their time and cooperated wholeheartedly. Their contributions have been indispensable for the data collection and analysis. The authors would like to express their sincere gratitude for the financial support from the following funds. This research was funded by the "New Generation of Artificial Intelligence" Major Project of Scientific and Technological Innovation 2030 (Grant No. 2023ZD0120600), the National Key R & D Program of China (Grant No. 2024YFB4504103), and the Key R & D Special Project of Henan Province (Grant No. 241111212300).

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